



[CSE 111s]

Logic Design

ELEVATOR

Project

Team 3 S2

Name	ID	Contribution	Precent
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Introduction:

We are required to design and implement a basic elevator system that operates with logic gates and sequential circuits.

The Elevator will be capable of transporting passengers between three floors (0,1,2)

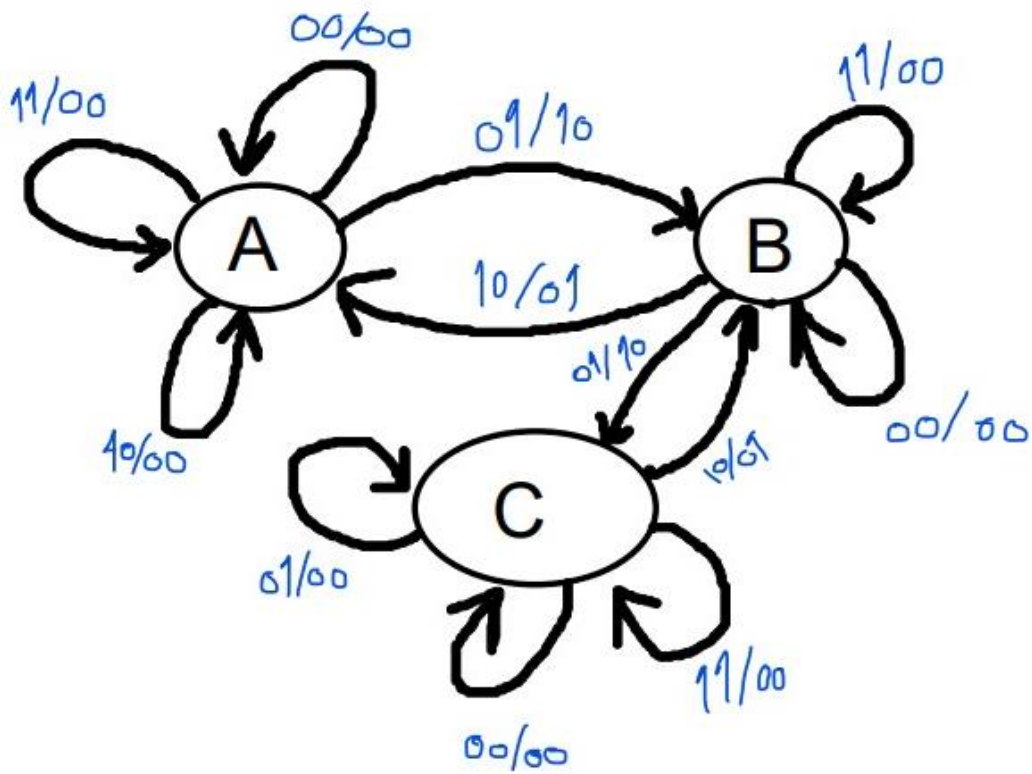
Components:

- Logic Gates: we needed various logic gates such as AND gates, OR gates, NOT gates, and XOR gates to implement the control logic for the elevator system.
- Sequential Circuits: we used sequential circuits, such as flip-flops, to store and manipulate the elevator's state and control signals.
- Input and Output Devices: we require input devices, such as buttons or switches, to signal the desired floor selection. Output devices, such as LEDs (Red one means you are moving up & Green Led means you are moving down) and (7-segment) will indicate the current floor (0 or 1 or 2).

The Main Components we used in this project:

- 1) 2 IC AND Gate (2 input 74LS08)
- 2) 1 IC OR Gate (2 input 74HC32)
- 3) 1 IC XOR Gate (2 input 74HC86)
- 4) 1 IC NOT Gate(74HC04)
- 5) 1 IC JK Flip Flop (74LS76)
- 6) 1 IC Timer (NE555P)
- 7) 1 IC Encoder (74LS47)
- 8) 7 segment display (Common Anode 5011BS)

State diagram:



We have 3 states: A,B,C

A= ground(0 level).

B= level 1

C= level 2

State table:

Assume (A=00,B=01,C=10,XX=11)

Present State	Next State				Output			
	x y				x y			
	00	01	10	11	00	01	10	11
A	A	B	A	A	00	10	00	00
B	B	C	A	B	00	10	01	00
C	C	C	B	C	00	00	01	00

Present State	Next State				Output			
	x y				x y			
	00	01	10	3	00	01	10	3
y1y2	Y1Y2				G R			
00	00	01	00	00	00	10	00	00
01	01	10	00	01	00	10	01	00
10	10	10	01	10	00	00	01	00
11	xx	xx	xx	xx	xx	xx	xx	xx

Jk Circuit's State Table:

Present State	Next State															
	xy															
	00				01				10				11			
	Y1Y2															
	J1 K1		J2K2		J1K1		J2K2		J1K1		J2K2		J1K1		J2K2	
00	0	x	0	x	0	x	1	x	0	x	0	x	0	x	0	x
01	0	x	x	0	1	x	x	1	0	x	x	1	0	x	x	0
10	x	0	0	x	x	0	0	x	x	1	1	x	x	0	0	x
11	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Equations:

$$1) J_1 = Y_2 \bar{x} y$$

$$2) K_1 = x \bar{y}$$

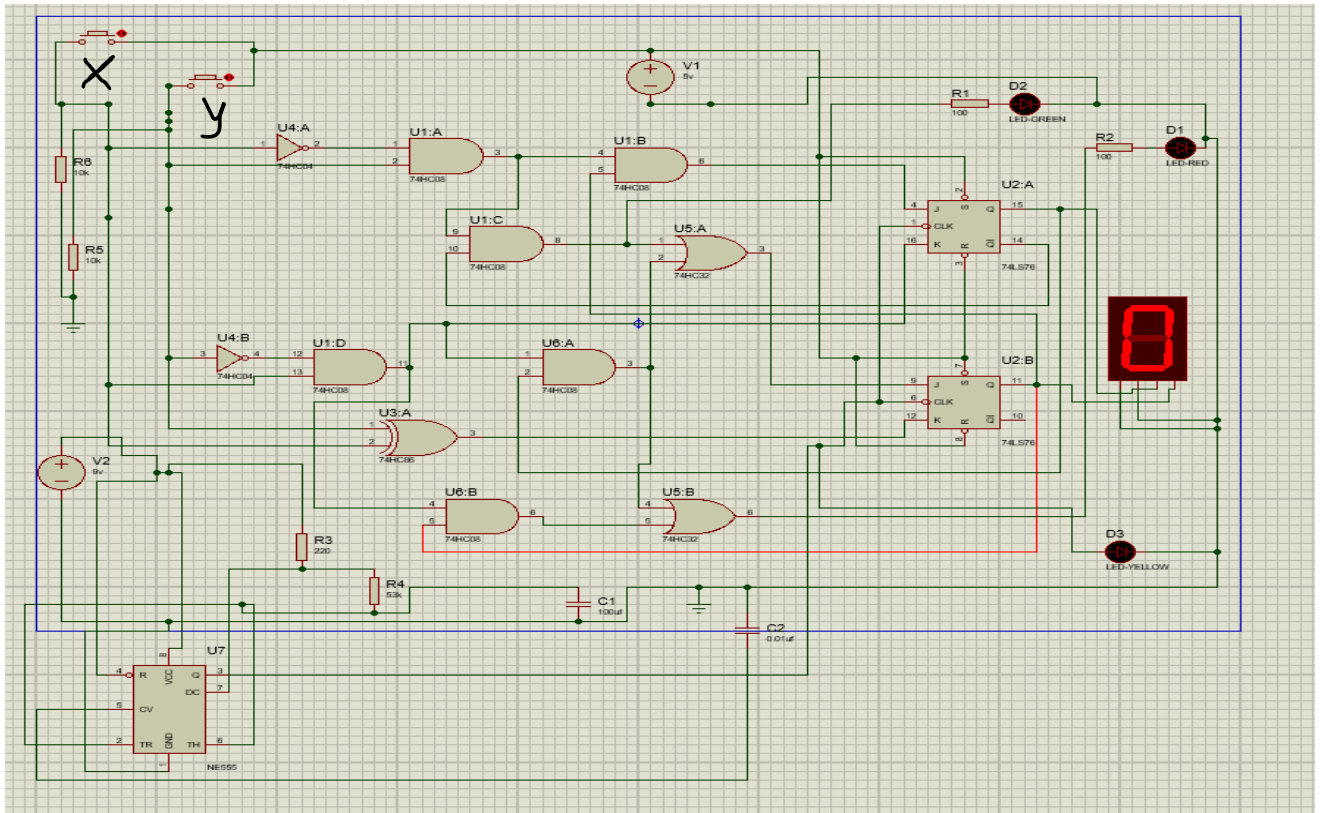
$$3) J_2 = \bar{x} y \bar{Y}_1 + x \bar{y} Y_1$$

$$4) K_2 = x \bar{y} + \bar{x} y = x \oplus y$$

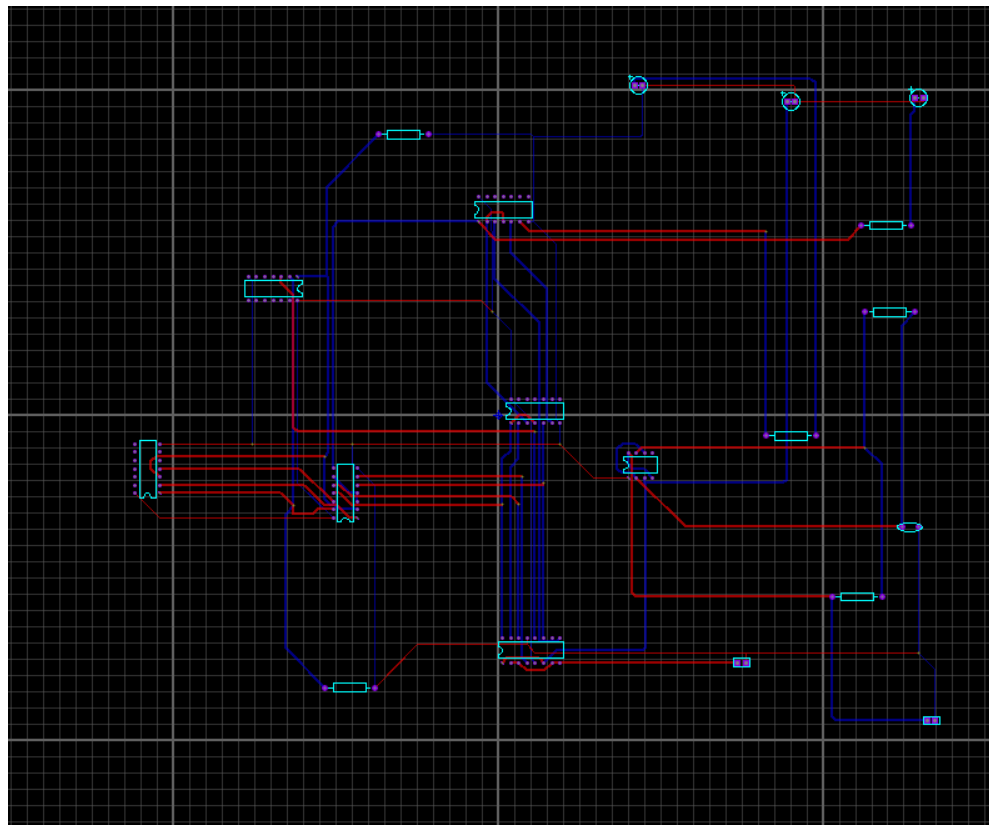
*We use 2 flip-flop type (JK-flip-flop) instead of 2 flip-flop type (D-flip-flop), Because of :

- D f/fs need more gates and complicated design than JK
- some components are broken down or has problems.

Circuit Diagram:



PCB double Layout:



555 Timer A stable Oscillator Circuit:

The output high-level duration T_H and low-level duration T_L can be calculated as follows:

$$R_A = 220 \text{ ohm}$$

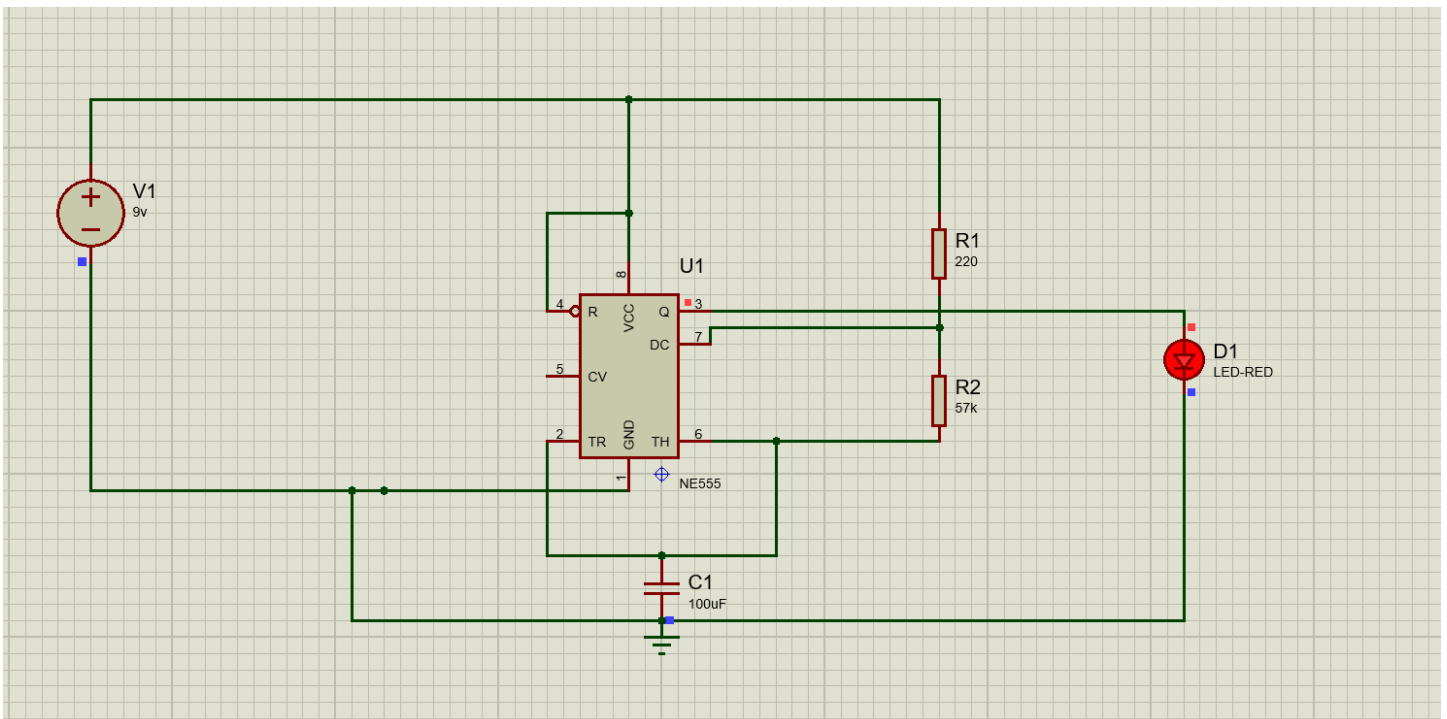
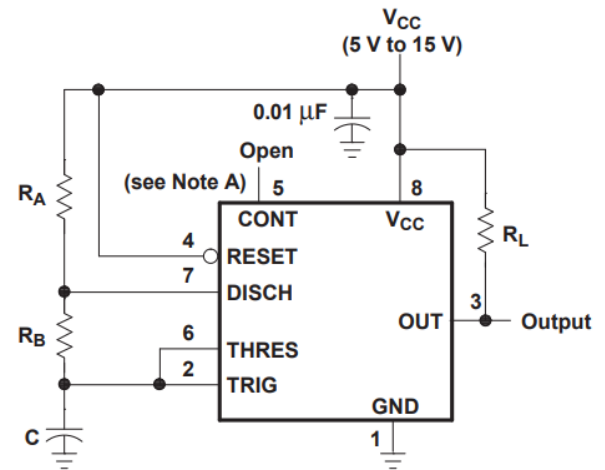
$$R_B = 57 \text{ k ohm}$$

$$C = 100 \mu\text{f}$$

$$T_H = 0.69 \times C \times (R_A + R_B) \sim 4 \text{ sec}$$

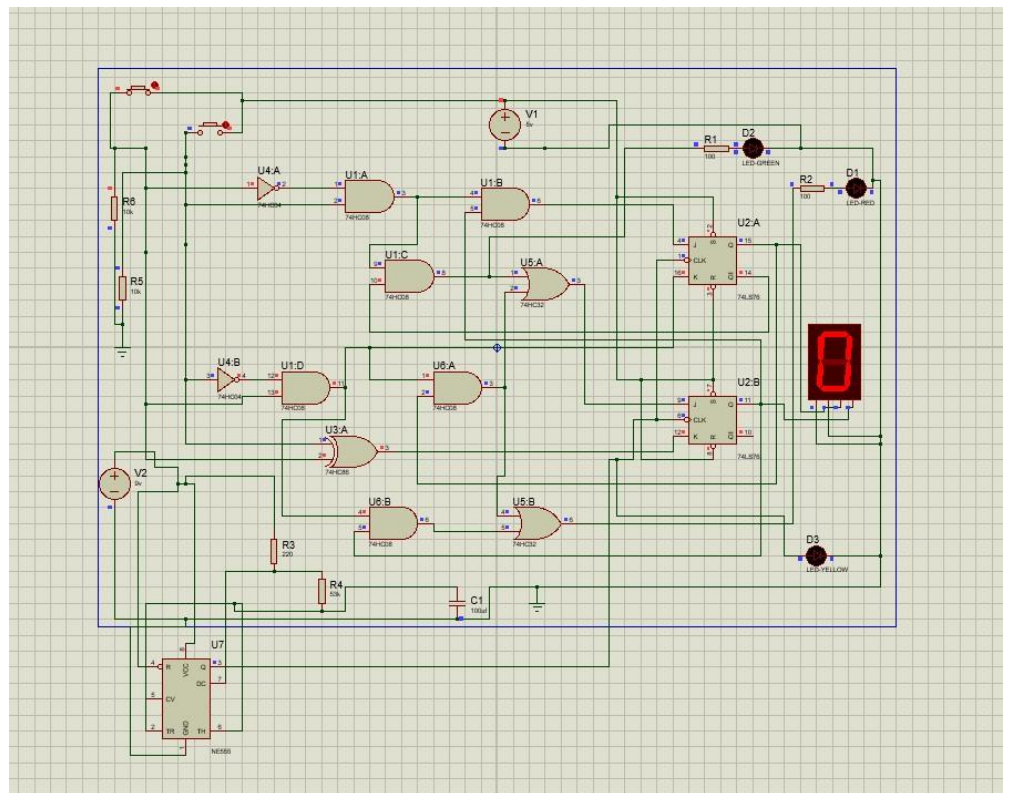
$$T_L = 0.69 \times C \times (R_B) \sim 4 \text{ sec}$$

$$T = T_H + T_L = 0.69 \times C \times (R_A + 2R_B) \sim 8 \text{ sec}$$

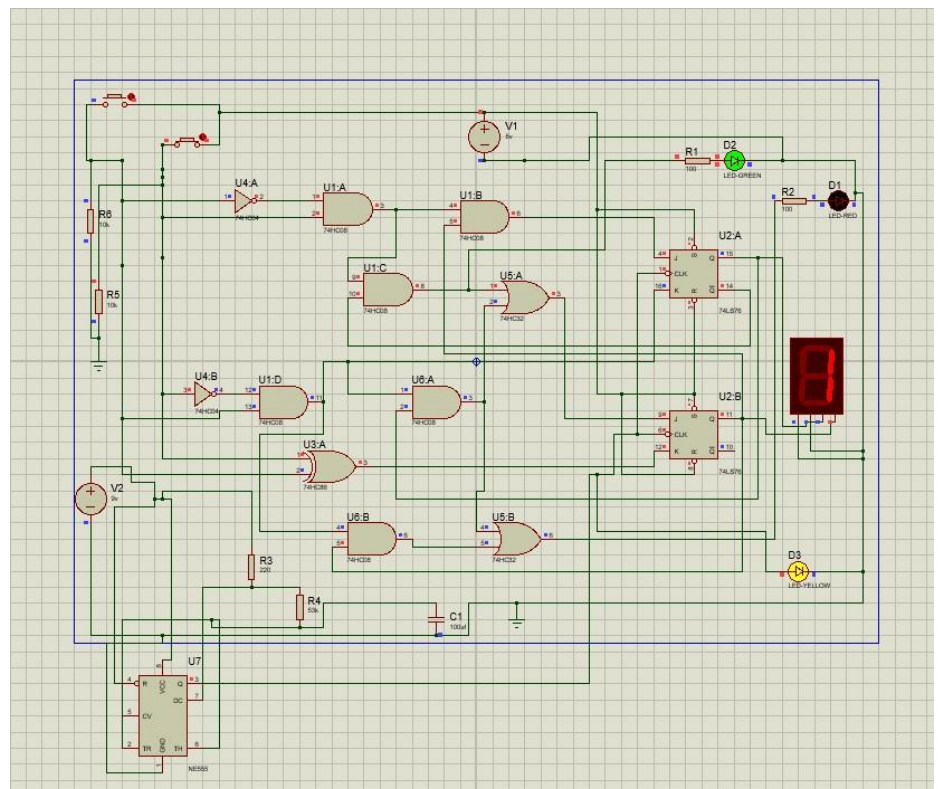


Simulation Results:

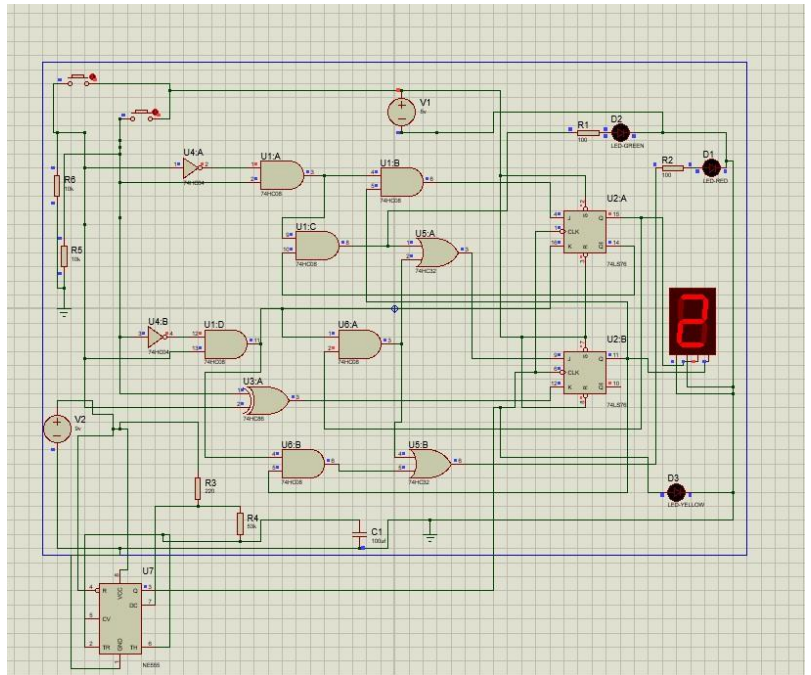
1st it is at 0 level



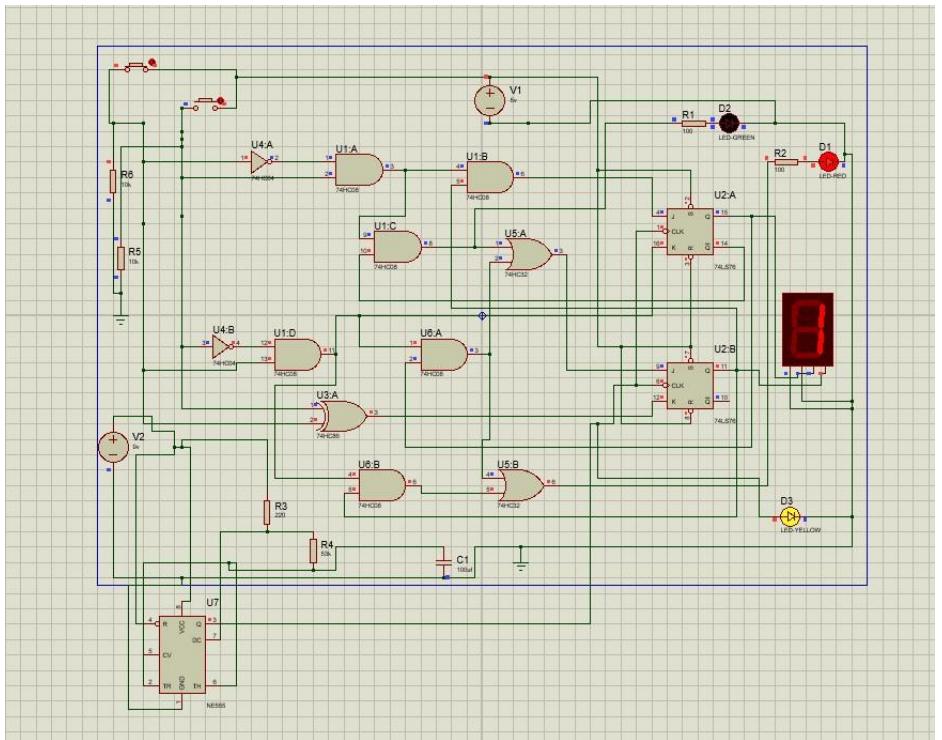
When the $xy=01$ And the clk is -ve edge , the green led will light until it reach level 1.



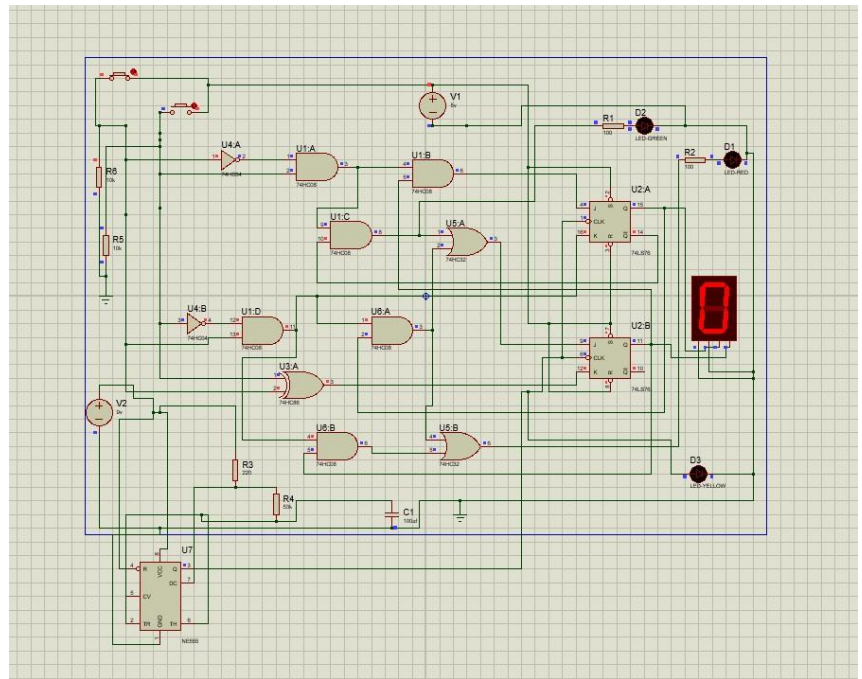
When the $xy=01$ And the clk is -ve edge and it is at level 1 , the green led will light until it reach level 2.



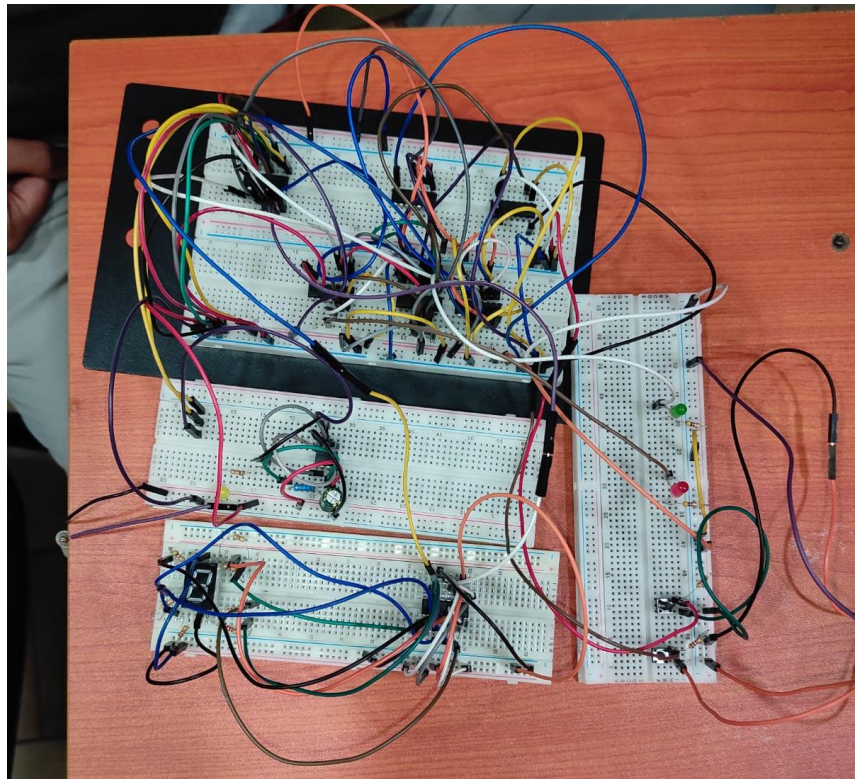
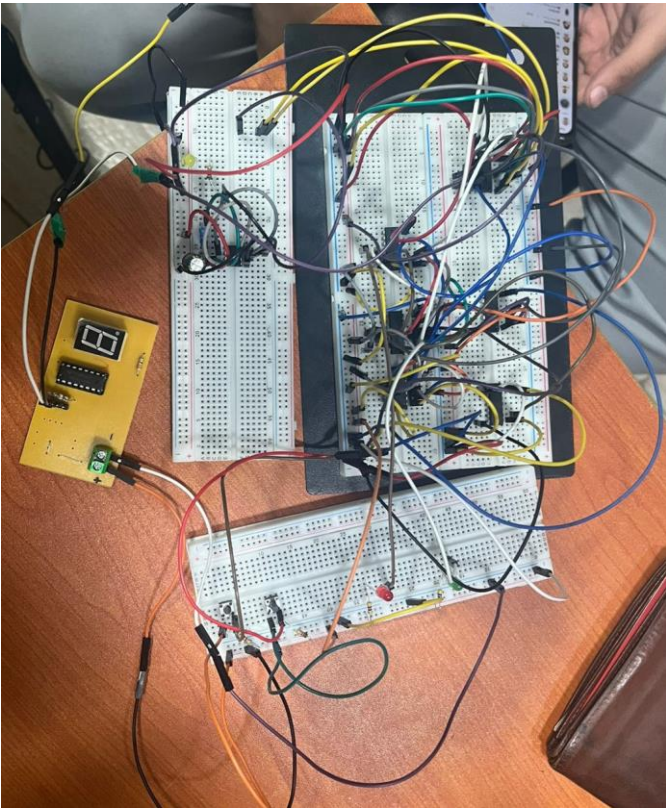
When the $xy=10$ And the clk is -ve edge and it is at level 2 , the red led will light until it reach level 1.



When the $xy=10$ And the clk is -ve edge and it is at level 1 , the red led will light until it reach level 0.



a physical prototype of the elevator system using the designed circuit and components:



Drive Links:

- 1- Video of simulation
- 2- File of simulation
- 3- Video for physical prototype

Link:

https://drive.google.com/drive/folders/1t0GrchqoJ9qULTbed2A8wqBnb_2yJL6f?usp=sharing